

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer.

Problem 1. (7 points) Find a particular solution y_p of the nonhomogeneous differential equation:

$$y'' - 2y' - 3y = 9x^2 + 1.$$

Step 1. Find the complementary solution $y_c(x)$.
characteristic equation:

$$r^2 - 2r - 3 = 0$$

$$(r-3)(r+1) = 0$$

$$r_1 = 3, r_2 = -1$$

$$\Rightarrow y_c(x) = C_1 e^{3x} + C_2 e^{-x}$$

Step 2. A particular solution $y_p(x)$ has a form of (Since $f(x) = 9x^2 + 1$)

$$y_p(x) = x^s (Ax^2 + Bx + C)$$

We can take $s=0$ to avoid any duplication in $y_c(x)$

$$y_p(x) = Ax^2 + Bx + C$$

$$y_p'(x) = 2Ax + B$$

$$y_p''(x) = 2A$$

put them into the equation. we get

$$[2A] - 2[2Ax + B] - 3[Ax^2 + Bx + C] = 9x^2 + 1$$

$$\Rightarrow \begin{cases} -3A = 9 \\ -4A - 3B = 0 \\ 2A - 2B - 3C = 1 \end{cases} \Rightarrow \begin{cases} A = -3 \\ B = 4 \\ C = -5 \end{cases}$$

$$\Rightarrow y_p(x) = -3x^2 + 4x - 5$$

Problem 2. (3 points) Based on the complementary solution of Problem 1, determine the appropriate form for a particular solution y_p of the nonhomogeneous differential equation:

$$y'' - 2y' - 3y = xe^{-x}.$$

$$f(x) = xe^{-x} \Rightarrow y_p(x) = x^s (Ax + B)e^{-x}$$

$$= Ax^{s+1}e^{-x} + Bx^s e^{-x}$$

To avoid any duplication of $y_c(x)$, we need take $s=1 \Rightarrow$

$$y_p(x) = Ax^2 e^{-x} + Bx e^{-x} = x(Ax + B)e^{-x}$$